

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab4/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/CIP-B103-Lab4

(kali㉿kali)-[~/CIP-B103-Lab4]
$ pwd
/home/kali/CIP-B103-Lab4

(kali㉿kali)-[~/CIP-B103-Lab4]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.3 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [103 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [169 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Fetched 75.9 MB in 1min 13s (1,041 kB/s)
1750 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali㉿kali)-[~/CIP-B103-Lab4]
$ sudo apt install -y wireshark tshark python3
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

```

Upgrading:

```

Upgrading:
  libjs-sphinxdoc  libpython3-stdlib  python3-dev  python3-tk
  libpython3-dev   python3      python3-minimal  python3-venv

Summary:
  Upgrading: 8, Installing: 0, Removing: 0, Not Upgrading: 1742
  Download size: 62.7 kB / 153 kB
  Space needed: 244 kB / 23.5 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 libjs-sphinxdoc all 9.1.0-4 [45.7 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libpython3-stdlib amd64 3.13.9-3+b1 [8,348 B]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 python3-tk amd64 3.13.9-3+b1 [8,640 B]
Fetched 62.7 kB in 1s (42.5 kB/s)
(Reading database... 432423 files and directories currently installed.)
Preparing to unpack .../python3-venv_3.13.9-3+b1_amd64.deb...
Unpacking python3-venv (3.13.9-3+b1) over (3.13.9-3)...
Preparing to unpack .../libpython3-dev_3.13.9-3+b1_amd64.deb...
Unpacking libpython3-dev:amd64 (3.13.9-3+b1) over (3.13.9-3)...
Preparing to unpack .../python3-dev_3.13.9-3+b1_amd64.deb...
Unpacking python3-dev (3.13.9-3+b1) over (3.13.9-3)...
Preparing to unpack .../python3-minimal_3.13.9-3+b1_amd64.deb...

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ wget -O evidence/smtp.pcap \
'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'

--2026-08-01 08:42:34-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.11.240, 172.67.75.39, 104.26.10.240, ..
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap [following]
--2026-08-01 08:42:35-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====>] 27.20K  --.-KB/s    in 0.001s

2026-08-01 08:42:36 (36.9 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap

(kali㉿kali)-[~/CIP-B103-Lab4]
$ sha256sum evidence/smtp.pcap working/smtp_working.pcap \
| tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  working/smtp_working.pcap

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ capinfos evidence/smtp.pcap \
| tee reports/smtp_capinfos.txt
File name:          evidence/smtp.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060

```

```

Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate:    2,920 bytes/s
Data bit rate:     23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:            17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:              3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$

```

```
(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -q -z conv,tcp \
  | tee reports/tcp_conversations.txt
```

TCP Conversations
Filter:<No Filter>

Total	Relative	Duration	Frames	Bytes	Frames	Bytes
10.10.1.4:1470	Start	↔ 74.53.140.153:25	25	1,980 bytes	28	22 kB
53 24 kB	0.036986000	7.5777				

```
(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number \
  -e frame.time \
  -e ip.src \
  -e tcp.srcport \
  -e ip.dst \
  -e tcp.dstport \
  -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv
```

```
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      25      10.10.1.4      1470      S:
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authori
ze the use of this system to transport unsolicited, | and/or bulk e-mail.
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      1470      74.53.140.153      25      C:
EHLO GP
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      25      10.10.1.4      1470      S:
250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LO
GIN | STARTTLS | HELP
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4      1470      74.53.140.153      25      C:
AUTH LOGIN
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      25      10.10.1.4      1470      S:
334 VxNlcm5hbWU6
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4      1470      74.53.140.153      25      C:
User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      25      10.10.1.4      1470      S:
334 UGFzc3dvcmQ6
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4      1470      74.53.140.153      25      C:
Pass: cHVuamFiQDEyMw==
15     2009-10-05T02:06:09.613798000-0400      74.53.140.153      25      10.10.1.4      1470      S:
```

```
16     2009-10-05T02:06:09.614414000-0400      10.10.1.4      1470      74.53.140.153      25      C:
MAIL FROM: <gurpartap@patriots.in>
17     2009-10-05T02:06:09.956765000-0400      74.53.140.153      25      10.10.1.4      1470      S:
250 OK
18     2009-10-05T02:06:09.957250000-0400      10.10.1.4      1470      74.53.140.153      25      C:
RCPT TO: <raj_deol2002in@yahoo.co.in>
19     2009-10-05T02:06:10.319708000-0400      74.53.140.153      25      10.10.1.4      1470      S:
250 Accepted
20     2009-10-05T02:06:10.320203000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA
21     2009-10-05T02:06:10.661679000-0400      74.53.140.153      25      10.10.1.4      1470      S:
354 Enter message, ending with "." on a line by itself
22     2009-10-05T02:06:10.692743000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
23     2009-10-05T02:06:10.692786000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
24     2009-10-05T02:06:10.692804000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
25     2009-10-05T02:06:10.692823000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Window Full] C: DATA fragment, 1460 bytes
26     2009-10-05T02:06:10.695115000-0400      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140
.153      25      Destination unreachable (Fragmentation needed)
28     2009-10-05T02:06:10.695623000-0400      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140
```



```

29      2009-10-05T02:06:10.696248000-0400      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140
.153      25      Destination unreachable (Fragmentation needed)
30      2009-10-05T02:06:10.696634000-0400      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140
.153      25      Destination unreachable (Fragmentation needed)
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
44      2009-10-05T02:06:11.858316000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
45      2009-10-05T02:06:11.858334000-0400      10.10.1.4      1470      74.53.140.153      25      fro

```

```

52      2009-10-05T02:06:12.248789000-0400      74.53.140.153      25      10.10.1.4      1470      S:
250 OK id=1Mugho-0003Dg-Un
54      2009-10-05T02:06:14.763825000-0400      10.10.1.4      1470      74.53.140.153      25      C:
QUIT
56      2009-10-05T02:06:15.105467000-0400      74.53.140.153      25      10.10.1.4      1470      S:
221 xc90.websitewelcome.com closing connection

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ wc -l reports/smtp_packet_inventory.tsv \
| tee reports/smtp_packet_count.txt
36 reports/smtp_packet_inventory.tsv

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ grep -Ei 'HELO|EHLO|MAIL FROM|RCPT TO|DATA|QUIT' \
reports/smtp_packet_inventory.tsv \
| tee reports/smtp_command_summary.txt
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      1470      74.53.140.153      25      C:
EHLO GP
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4      1470      74.53.140.153      25      C:
MAIL FROM: <gurpartap@patriots.in>
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4      1470      74.53.140.153      25      C:
RCPT TO: <raj_deol2002in@yahoo.co.in>
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4      1470      74.53.140.153      25      C:

```

```

22      2009-10-05T02:06:10.692743000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
24      2009-10-05T02:06:10.692804000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1460 bytes
25      2009-10-05T02:06:10.692823000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Window Full] C: DATA fragment, 1460 bytes
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4      1470      74.53.140.153      25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4      1470      74.53.140.153      25      C:
DATA fragment, 1452 bytes
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4      1470      74.53.140.153      25      C:

```

```
(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e smtp.req.command \
-e smtp.req.parameter \
-e smtp.response.code \
-e smtp.response \
| tee reports/smtp_commands_responses.tsv
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      10.10.1.4      220
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      74.53.140.153      EHLO      GP      250
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      10.10.1.4      250-xc90.websitewelcome.com Hello GP [122.162.143.157]\r\n
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4      74.53.140.153      AUTH      LOGIN      334
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      10.10.1.4      334 VxNlcm5hbWU6\r\n
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4      74.53.140.153
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      10.10.1.4      334
334 UGFzc3dvcmQ6\r\n
```

```
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4      74.53.140.153
15     2009-10-05T02:06:09.613798000-0400      74.53.140.153      10.10.1.4      235
235 Authentication succeeded\r\n
16     2009-10-05T02:06:09.614414000-0400      10.10.1.4      74.53.140.153      MAIL      FROM: <gurp
artap@patriots.in>
17     2009-10-05T02:06:09.956765000-0400      74.53.140.153      10.10.1.4      250
250 OK\r\n
18     2009-10-05T02:06:09.957250000-0400      10.10.1.4      74.53.140.153      RCPT      TO: <raj_de
ol2002in@yahoo.co.in>
19     2009-10-05T02:06:10.319708000-0400      74.53.140.153      10.10.1.4      250
250 Accepted\r\n
20     2009-10-05T02:06:10.320203000-0400      10.10.1.4      74.53.140.153      DATA
21     2009-10-05T02:06:10.661679000-0400      74.53.140.153      10.10.1.4      354
354 Enter message, ending with "." on a line by itself\r\n
52     2009-10-05T02:06:12.248789000-0400      74.53.140.153      10.10.1.4      250
250 OK id=1Mugho-0003Dg-Un\r\n
54     2009-10-05T02:06:14.763825000-0400      10.10.1.4      74.53.140.153      QUIT
56     2009-10-05T02:06:15.105467000-0400      74.53.140.153      10.10.1.4      221
221 xc90.websitewelcome.com closing connection\r\n
```

```
(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y smtp -T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e smtp.req.command \
-e smtp.req.parameter \
-e smtp.response.code \
-e smtp.response
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      10.10.1.4      220
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4      74.53.140.153      EHLO      GP      250
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      10.10.1.4      250-xc90.websitewelcome.com Hello GP [122.162.143.157]\r\n
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4      74.53.140.153      AUTH      LOGIN      334
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      10.10.1.4      334 VxNlcm5hbWU6\r\n
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4      74.53.140.153
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      10.10.1.4      334
334 UGFzc3dvcmQ6\r\n
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4      74.53.140.153
```

```

15 2009-10-05T02:06:09.613798000-0400 74.53.140.153 10.10.1.4 235
235 Authentication succeeded\r\n
16 2009-10-05T02:06:09.614414000-0400 10.10.1.4 74.53.140.153 MAIL FROM: <gurp
artap@patriots.in>
17 2009-10-05T02:06:09.956765000-0400 74.53.140.153 10.10.1.4 250
250 OK\r\n
18 2009-10-05T02:06:09.957250000-0400 10.10.1.4 74.53.140.153 RCPT TO: <raj_de
ol2002in@yahoo.co.in>
19 2009-10-05T02:06:10.319708000-0400 74.53.140.153 10.10.1.4 250
250 Accepted\r\n
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 74.53.140.153 DATA
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 10.10.1.4 354
354 Enter message, ending with "." on a line by itself\r\n
22 2009-10-05T02:06:10.692743000-0400 10.10.1.4 74.53.140.153
23 2009-10-05T02:06:10.692786000-0400 10.10.1.4 74.53.140.153
24 2009-10-05T02:06:10.692804000-0400 10.10.1.4 74.53.140.153
25 2009-10-05T02:06:10.692823000-0400 10.10.1.4 74.53.140.153
26 2009-10-05T02:06:10.695115000-0400 192.168.1.1,10.10.1.4 10.10.1.4,74.53.140.153
28 2009-10-05T02:06:10.695623000-0400 192.168.1.1,10.10.1.4 10.10.1.4,74.53.140.153
29 2009-10-05T02:06:10.696248000-0400 192.168.1.1,10.10.1.4 10.10.1.4,74.53.140.153
30 2009-10-05T02:06:10.696634000-0400 192.168.1.1,10.10.1.4 10.10.1.4,74.53.140.153

32 2009-10-05T02:06:11.104972000-0400 10.10.1.4 74.53.140.153
33 2009-10-05T02:06:11.104998000-0400 10.10.1.4 74.53.140.153
35 2009-10-05T02:06:11.469759000-0400 10.10.1.4 74.53.140.153
36 2009-10-05T02:06:11.469814000-0400 10.10.1.4 74.53.140.153
38 2009-10-05T02:06:11.494181000-0400 10.10.1.4 74.53.140.153
39 2009-10-05T02:06:11.494199000-0400 10.10.1.4 74.53.140.153
41 2009-10-05T02:06:11.834628000-0400 10.10.1.4 74.53.140.153
42 2009-10-05T02:06:11.834655000-0400 10.10.1.4 74.53.140.153
44 2009-10-05T02:06:11.858316000-0400 10.10.1.4 74.53.140.153
45 2009-10-05T02:06:11.858334000-0400 10.10.1.4 74.53.140.153
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 10.10.1.4 250
250 OK id=1Mugho-0003Dg-Un\r\n
54 2009-10-05T02:06:14.763825000-0400 10.10.1.4 74.53.140.153 QUIT
56 2009-10-05T02:06:15.105467000-0400 74.53.140.153 10.10.1.4 221
221 xc90.websitewelcome.com closing connection\r\n

(kali㉿kali)-[~/CIP-B103-Lab4]

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y "frame.number==12 || frame.number==14" -x
0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 ..3..`...<....E.
0010 00 46 25 29 40 00 80 06 f3 ac 0a 0a 01 04 4a 35 .F%)@.....J5
0020 8c 99 05 be 00 19 7e c4 53 c6 ae ec 63 00 50 18 .....~.S...c.P.
0030 fe af 22 a4 00 00 5a 33 56 79 63 47 46 79 64 47 .."...Z3VycGFydG
0040 46 77 51 48 42 68 64 48 4a 70 62 33 52 7a 4c 6d FwQHBhdHJpb3RzLm
0050 6c 75 0d 0a lu..

0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 ..3..`...<....E.
0010 00 3a 25 2e 40 00 80 06 f3 b3 0a 0a 01 04 4a 35 .:%.@.....J5
0020 8c 99 05 be 00 19 7e c4 53 e4 ae ec 63 12 50 18 .....~.S...c.P.
0030 fe 9d 54 b1 00 00 63 48 56 75 61 6d 46 69 51 44 ..T...cHVuamFiQD
0040 45 79 4d 77 3d 3d 0d 0a EyMw==..

(kali㉿kali)-[~/CIP-B103-Lab4]
$ python3 - <<'PYLAB'
import base64

samples = {
    'username': 'THE_BASE64_USERNAME_FROM_FRAME_12',

```



```

    'password': 'THE_BASE64_PASSWORD_FROM_FRAME_14'
}

for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')

PYLAB
username: Lq!:♦D♦♦
                        TN0T@0Mv
password: Lq!:♦♦Ic♦♦
                        TN0T@0Mx

(kali㉿kali)-[~/CIP-B103-Lab4]
$ python3 - <<'PYLAB'
import base64

samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',

```

```

                        TN0T@0Mx

(kali㉿kali)-[~/CIP-B103-Lab4]
$ python3 - <<'PYLAB'
import base64

samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}

for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')

PYLAB
username: gurpartap@patriots.in
password: punjab@123

(kali㉿kali)-[~/CIP-B103-Lab4]
$

```

```
(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y smtp -T fields -e tcp.stream | sort -u
0

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y "smtp.req.command=AUTH" -T fields -e tcp.stream
0

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
```

```
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLm1u

18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

30
235 Authentication succeeded

36
```



```

30
235 Authentication succeeded

36
MAIL FROM: <gurpartap@patriots.in>

8
250 OK

39
RCPT TO: <raj_deol2002in@yahoo.co.in>

14
250 Accepted

6
DATA

56
354 Enter message, ending with "." on a line by itself

1460

```

```

(kali㉿kali)-[~/CIP-B103-Lab4]
$ grep -Ei '^ (Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtplib_stream_0.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(kali㉿kali)-[~/CIP-B103-Lab4]
$ grep -A50 "DATA" reports/smtplib_stream_0.txt
DATA

```

```

354 Enter message, ending with "." on a line by itself

1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
  .boundary="====_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjZEDer1Kh8i+hUyVo0A=
Content-Language: en-us
x-cr-hashedpuzzle: SeA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ JO7s Kum6 KytW LFcI LjUt;
1;cgBhAGoAXwBkAGUAbwBsADIAAAwADIAAQBuAEAAeQBhAGGAbwBvAC4AYwBvAC4AaQBuAA=;Sosha1_v1;7;{CAA37F59-18
50-45C7-8540-AA27696B5398};ZwB1AHIAcABhAHIAcABhAHAAQABwAGEAdABYAGkAbwB0AHMALgBpAG4A;Mon, 05 Oct 200
9 06:06:01 GMT;UwBNAFQAUA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

```

```
-----=_NextPart_000_0004_01CA45B0.095693F0
Content-Type: multipart/alternative;
.boundary="-----=_NextPart_001_0005_01CA45B0.095693F0"
```

```
-----=_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/plain;
.charset="us-ascii"
Content-Transfer-Encoding: 7bit
```

Hello

I send u smtp pcap file

Find the attachment

GPS

GPS

```
-----=_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/html;
.charset="us-ascii"
Content-Transfer-Encoding: quoted-printable
```

<html xmlns:v=3D"urn:schemas-microsoft-com:vm" =

```
(kali@kali)-[~/CIP-B103-Lab4]
$
```

```
(kali@kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number \
  -e frame.time_epoch \
  -e eth.src \
  -e eth.dst \
  -e ip.src \
  -e tcp.srcport \
  -e ip.dst \
  -e tcp.dstport \
  -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv
6      1254722768.219663000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      251
0.10.1.4      1470      0
7      1254722768.224809000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      147
0      74.53.140.153      25      0
9      1254722768.566183000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      251
0.10.1.4      1470      0
10     1254722768.568729000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      147
0      74.53.140.153      25      0
11     1254722768.911081000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      251
0.10.1.4      1470      0
```

0	74.53.140.153	25	0			
13	1254722769.253544000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
14	1254722769.254118000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
15	1254722769.613798000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
16	1254722769.614414000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
17	1254722769.956765000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
18	1254722769.957250000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
19	1254722770.319708000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
20	1254722770.320203000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
21	1254722770.661679000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
22	1254722770.692743000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
23	1254722770.692786000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			

0	74.53.140.153	25	0			
25	1254722770.692823000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
26	1254722770.695115000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	192.168.1.1,10.10.1	
.4	1470	10.10.1.4,74.53.140.153	25	0		
28	1254722770.695623000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	192.168.1.1,10.10.1	
.4	1470	10.10.1.4,74.53.140.153	25	0		
29	1254722770.696248000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	192.168.1.1,10.10.1	
.4	1470	10.10.1.4,74.53.140.153	25	0		
30	1254722770.696634000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	192.168.1.1,10.10.1	
.4	1470	10.10.1.4,74.53.140.153	25	0		
32	1254722771.104972000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
33	1254722771.104998000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
35	1254722771.469759000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
36	1254722771.469814000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
38	1254722771.494181000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
39	1254722771.494199000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			

0	74.53.140.153	25	0			
42	1254722771.834655000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
44	1254722771.858316000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
45	1254722771.858334000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
52	1254722772.248789000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				
54	1254722774.763825000		00:e0:1c:3c:17:c2	00:1f:33:d9:81:60	10.10.1.4	147
0	74.53.140.153	25	0			
56	1254722775.105467000		00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153	251
0.10.1.4	1470	0				

```

(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap \
-Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields \
-e frame.number \
-e tcp.stream \
-e smtp.req.parameter \
-e data-text-lines \
| tee reports/client_indicators.tsv

```

```
-Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields \
-e frame.number \
-e tcp.stream \
-e smtp.req.parameter \
-e data-text-lines \
| tee reports/client_indicators.tsv
22      0      Line-based text data (47 lines)
26      0      Line-based text data (12 lines)

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y smtp | grep -i STARTTLS
  9      1.074123 74.53.140.153 → 10.10.1.4      SMTP 191 S: 250-xc90.websitewelcome.com Hello GP [122
.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y "smtp.req.command=STARTTLS"

(kali㉿kali)-[~/CIP-B103-Lab4]
$ tshark -r working/smtp_working.pcap -Y tls

(kali㉿kali)-[~/CIP-B103-Lab4]
$
```